BIOL-1 Practice Test 02 — Answer Sheet

Exam 02 Preparation (Modules 7–11)

Name:		Date:		Score:	

✂️ **TEAR-OFF ANSWER SHEET:** record every Part A: Multiple Choice (25 questions) and Part B: Fill in the Blank (10 questions) answer on this page. Free response begins on the reverse side. Detach and turn in this sheet.

Part A: Multiple Choice (25 questions)

1.		**10.**		**19.**	
2.		**11.**		**20.**	
3.		**12.**		**21.**	
4.		**13.**		**22.**	
5.		**14.**		**23.**	
6.		**15.**		**24.**	
7.		**16.**		**25.**	
8.		**17.**			
9.		**18.**			

Part B: Fill in the Blank (10 questions)

1.		**6.**	
2.		**7.**	
3.		**8.**	
4.		**9.**	
5.		**10.**	

BIOL-1 Practice Test 02 — Part C: Short Answer (4 questions)

Free response. Write clearly in the lined spaces.

1. **(Module 7)** Write the complementary DNA sequence AND the transcribed mRNA sequence for the following DNA template strand: **3′-TAC GGG TTA ACG ACT-5′**.
2. **(Module 8 & 9)** Explain the difference between Mitosis and Meiosis in terms of their overall purpose for a multicellular organism like a human. Briefly link how Meiosis connects to Mendel's Law of Segregation.
3. **(Module 10)** Describe how it is possible for a muscle cell and a nerve cell in your body to look and function so differently, despite containing the exact same DNA.
4. **(Module 11)** Choose ONE biotechnology tool discussed in class (e.g., PCR, Gel Electrophoresis, CRISPR, Recombinant Plasmids). Explain what it does and provide one real-world application or example of its use.

Response 1 — Question Number: ______

Response 2 — Question Number: ______

Response 3 — Question Number: ______

Response 4 — Question Number: ______

BIOL-1 Practice Test 02 — Question Booklet

Answer Parts A and B on the tear-off sheet. Keep this booklet with your exam materials.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 7: Molecular Genetics

1. The enzyme that unwinds the DNA double helix during replication is:
    - A) RNA polymerase
    - B) DNA polymerase
    - C) Helicase
    - D) Ligase
2. Okazaki fragments are synthesized on the:
    - A) Lagging strand
    - B) mRNA transcript
    - C) Leading strand
    - D) tRNA molecule
3. Transcription is the process of converting:
    - A) RNA into a protein
    - B) DNA into RNA
    - C) DNA into a new DNA molecule
    - D) Protein into an observable trait
4. Which part of a primary mRNA transcript is removed during RNA splicing before it leaves the nucleus?
    - A) Introns
    - B) Promoters
    - C) Codons

◦ D) Exons

5. A mutation that shifts the entire reading frame of the ribosome is called a:

    ◦ A) Missense mutation
    ◦ B) Silent mutation
    ◦ C) Frameshift mutation
    ◦ D) Point mutation

Module 8: Cellular Genetics (Mitosis & Meiosis)

1. Which phase of the cell cycle is characterized by DNA replication?

    ◦ A) G2 phase
    ◦ B) M phase
    ◦ C) S phase
    ◦ D) G1 phase

2. After a single human cell undergoes meiosis, the resulting cells are:

    ◦ A) 4 haploid, genetically distinct cells
    ◦ B) 2 diploid, genetically identical cells
    ◦ C) 4 diploid, genetically distinct cells
    ◦ D) 2 haploid, genetically identical cells

3. Crossing over, which increases genetic diversity, occurs during:

    ◦ A) Metaphase II of Meiosis
    ◦ B) Prophase of Mitosis
    ◦ C) Prophase I of Meiosis
    ◦ D) Anaphase I of Meiosis

4. Nondisjunction during meiosis can lead to conditions such as Down syndrome, which is caused by trisomy of chromosome:

    ◦ A) 18
    ◦ B) 21

◦ C) 23
◦ D) Y

5. The p53 protein is heavily involved in:

◦ A) Forming the cleavage furrow
◦ B) Breaking down the nuclear envelope
◦ C) Stopping the cell cycle if DNA is damaged
◦ D) Attaching spindle fibers to kinetochores

Module 9: Inheritance Genetics

1. The physical appearance or observable trait of an organism is its:

◦ A) Genotype
◦ B) Phenotype
◦ C) Karyotype
◦ D) Allele

2. In a monohybrid cross between two heterozygous parents (Aa × Aa), what is the expected phenotypic ratio for a completely dominant trait?

◦ A) 1:2:1
◦ B) 3:1
◦ C) 9:3:3:1
◦ D) 1:1

3. A true-breeding red snapdragon is crossed with a true-breeding white snapdragon, producing offspring that are all pink. This is an example of:

◦ A) Incomplete dominance
◦ B) Polygenic inheritance
◦ C) Sex-linked inheritance
◦ D) Codominance

4. Human blood types exhibit which type(s) of inheritance pattern(s)?
    - A) Sex-linked recessive
    - B) Simple dominance only
    - C) Incomplete dominance
    - D) Codominance and multiple alleles
5. Why are sex-linked recessive traits more common in human males than females?
    - A) Males have a stronger Y chromosome.
    - B) The X chromosome in males is larger than in females.
    - C) Males only have one X chromosome, so a single recessive allele determines the phenotype.
    - D) Females lack the genes required for these traits entirely.

Module 10: Epigenetics

1. Epigenetics refers to changes in gene expression that:
    - A) Only occur in prokaryotes.
    - B) Do not change the DNA sequence but can be passed to daughter cells.
    - C) Alter the underlying DNA sequence permanently.
    - D) Are only caused by environmental toxins.
2. DNA methylation generally has what effect on gene expression?
    - A) It increases transcription.
    - B) It causes the DNA to mutate rapidly.
    - C) It has no effect on transcription.
    - D) It decreases or silences transcription.
3. The lac operon in E. coli allows the bacteria to produce enzymes to digest lactose only when lactose is present. This is an example of:
    - A) A sex-linked trait
    - B) A frameshift mutation
    - C) Epigenetic inheritance

◦ D) Gene regulation

4. Acetylation of histones usually results in:

   ◦ A) Loosely packed chromatin and increased transcription.
   ◦ B) Immediate degradation of the DNA.
   ◦ C) Tightly packed chromatin and decreased transcription.
   ◦ D) The splicing of introns out of mRNA.

5. The random inactivation of one X chromosome in female mammalian cells results in physical traits like tortoiseshell coloring in cats. The inactivated X chromosome condenses into a structure known as a:

   ◦ A) Centrosome
   ◦ B) Barr body
   ◦ C) Telomere
   ◦ D) Nucleosome

Module 11: Genomics & Biotechnology

1. The primary function of restriction enzymes in biotechnology is to:

   ◦ A) Amplify DNA sequences.
   ◦ B) Analyze protein folding.
   ◦ C) Glue Okazaki fragments together.
   ◦ D) Cut DNA at highly specific sequence sites.

2. Polymerase Chain Reaction (PCR) is used to:

   ◦ A) Rapidly make millions of copies of a specific DNA segment.
   ◦ B) Translate mRNA into a polypeptide chain in a test tube.
   ◦ C) Separate DNA fragments by size.
   ◦ D) Insert foreign DNA into a bacterial plasmid.

3. During gel electrophoresis, DNA fragments move toward the positive electrode because:

   ◦ A) DNA is positively charged.

- ◦ B) DNA has a neutral charge but is pulled by gravity.
- ◦ C) agarose gel attracts DNA chemically.
- ◦ D) the phosphate backbone gives DNA a negative charge.

4. The CRISPR-Cas9 system was originally discovered in:
   - ◦ A) Bacteria as a rudimentary cellular immune system against viral bacteriophages.
   - ◦ B) Plants to resist fungal infections.
   - ◦ C) Human liver cells to repair damaged DNA.
   - ◦ D) Viruses as an attack mechanism against hosts.
5. Transgenic organisms, such as bacteria engineered to produce human insulin, are possible because:
   - ◦ A) Transcription only works in eukaryotic laboratory cells.
   - ◦ B) Bacteria and humans have identical DNA sequences.
   - ◦ C) Human genes naturally mutate into bacterial genes.
   - ◦ D) All organisms share a nearly universal genetic code.

Part B: Fill in the Blank (10 questions)

Word Bank: Biotechnology, Cytokinesis, Epigenetic, Heterozygous, Lethal, mRNA, Operon, Smaller, Trisomy, Uracil

1. In RNA, the nitrogenous base ____ replaces the Thymine found in DNA.
2. According to the Central Dogma of molecular biology, DNA is transcribed into ____, which is then translated into a protein.
3. The division of the cytoplasm at the very end of mitosis or meiosis is called ____.
4. The presence of three copies of a chromosome instead of the normal two is referred to as a ____.

5. A gene variant that causes death when inherited in homozygous form is known as a ____ allele.

6. An individual with two different alleles for a specific trait (e.g., Bb) is called ____.

7. In gene regulation, a(n) ____ is a cluster of genes under the control of a single promoter, commonly found in prokaryotes.

8. Identical twins have the exact same DNA sequence, but as they age, differences in their gene expression accumulate due to ____ changes.

9. In gel electrophoresis, ____ fragments of DNA travel through the gel matrix faster and farther than larger fragments.

10. The use of an organism, or a component of an organism, to make a product or process for human use is known as ____.